

EXPT NO:6	INTEGRATE DATA FROM VARIOUS SOURCES FOR COMPREHENSIVE ANALYSIS USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is data integration in the context of Business Intelligence (BI)?

Data integration in Business Intelligence (BI) is the process of combining data from multiple sources into a single, unified dataset for analysis. It helps organizations obtain a complete view of business operations, improve data consistency, and make better decisions based on comprehensive information.

2. Explain the difference between a join and a data blend in Tableau.

A join combines data from two or more tables into a single table based on a common field before analysis. A data blend combines data from different data sources at the visualization level, where one data source acts as the primary source and the other as the secondary source. Joins are generally used when the datasets are closely related, while data blending is useful when the data comes from different sources that cannot be directly joined.

3. What are relationships and cardinality (e.g., one-to-many, many-to-many) in Power BI's data model?

Relationships in Power BI connect tables using a common field so that data from multiple tables can be analyzed together. Cardinality defines how records are related between tables. A one-to-many relationship means one record in one table is related to multiple records in another table, while a many-to-many relationship means multiple records in both tables are related to each other.

4. Why is data quality (e.g., consistent naming conventions, data types) crucial when integrating data from different sources?

High-quality data is essential because inconsistent region names, incorrect data types, duplicate records, or missing values can lead to incorrect relationships and inaccurate analysis. Maintaining consistent naming conventions and proper data types ensures reliable calculations and meaningful visualizations.

5. Name two common challenges encountered during data integration.

Two common challenges are:

- Inconsistent data formats or naming conventions between different data sources.
- Missing, duplicate, or incorrect data that can affect the accuracy of the integrated dataset and analysis.

IN-LAB EXERCISE:

OBJECTIVE:

To connect to, combine, and model data from multiple disparate sources in Tableau and Power BI to create a unified, comprehensive dashboard for holistic analysis. PROCEDURE:

1. Data Collection:

- Import data from at least two different sources (e.g., a CSV file and an Excel file).
- Dataset 1: Sales Data (CSV file) with fields like Order ID, Order Date, Region, Sales Amount.
- Dataset 2: Regional Sales Targets (Excel file) with fields like Region, Sales Target, Target Year.

2. Data Preparation and Transformation:

- Clean both datasets: Check for inconsistencies in region names (e.g., "N. America" vs. "North America").
- Ensure data types are correct (e.g., Sales Amount is numeric, Order Date is a date).
- Handle any missing values as appropriate.

3. Data Integration and Modeling:

- Tableau: Use the Data Source pane to establish a relationship or join between the Sales Data and Regional Targets tables using the common Region field.
- Power BI: Use the Model view to create a relationship between the two tables, connecting them on the Region column. Define the cardinality (e.g., many-to-one).

4. Create Calculated Fields for Comparison:

- Develop new measures that leverage both datasets.
- Examples: $\text{Target Achievement \%} = (\text{SUM}(\text{Sales}) / \text{SUM}(\text{Sales Target})) * 100$
- $\text{Sales vs. Target Variance} = \text{SUM}(\text{Sales}) - \text{SUM}(\text{Sales Target})$

5. Design a Comprehensive Dashboard:

- Create visuals that combine data from both sources to provide a single view of performance.
- Add KPI cards for Total Sales and Overall Target Achievement %.
- Use a gauge chart or bullet chart to show sales performance against targets for each region.

- Create a combined bar chart showing Actual Sales vs. Target Sales by Region.
- Use a map to visualize the Sales vs. Target Variance, highlighting underperforming or overperforming regions.

6. Validate Data Integrity:

- Verify that the joined/related data is accurate by cross-checking values.
- Ensure filters on the dashboard correctly update visuals that use data from both sources.

7. Publishing and Sharing:

- Tableau: Publish the workbook to Tableau Public or Tableau Server.
- Power BI: Publish the report to Power BI Service and share the link.

Experiment Datasets:

Dataset 1: Retail Sales Data (CSV) Fields

Used:

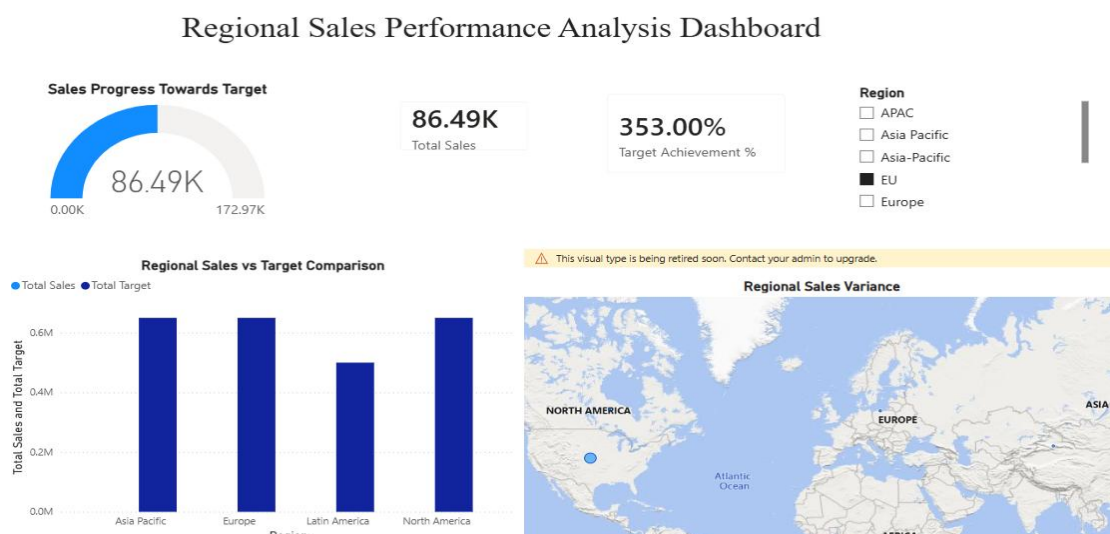
- Order ID – Unique identifier for each order
- Order Date – Date of the sales transaction
- Region – Sales region
- Sales – Sales revenue from the order

Dataset 2: Regional Sales Targets (Excel) Fields

Used:

- Region – Sales region (used as the key for joining/relationship)
- Sales Target – The sales goal for the region
- Target Year – The year for which the target is set

OUTPUT :



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

- 1. How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?**

Integrating the sales targets data allowed actual sales to be compared with planned sales targets. This made it possible to evaluate regional performance, calculate target achievement percentages, identify underperforming regions, and perform variance analysis, which would not be possible using sales data alone.

- 2. Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.**

A many-to-one relationship was created between the Sales_Data table and the Regional_Sales_Targets table using the Region field. This relationship was appropriate because multiple sales records belong to a single region, while each region has one corresponding sales target. It enabled accurate comparison of actual sales with target values.

- 3. What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?**

The main challenge was inconsistent region names between the datasets, such as abbreviated and full region names (e.g., N. America and North America). These inconsistencies had to be corrected before creating relationships to ensure accurate data integration and analysis.

- 4. How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?**

Variance calculations help managers compare actual sales with target sales. They quickly identify regions that are exceeding or falling short of their targets, allowing managers to take corrective actions, allocate resources effectively, and improve overall business performance.

- 5. What other data source (e.g., marketing campaign data, customer demographics) could you integrate to further enrich this analysis, and what new insights might it provide?**

Additional data sources such as marketing campaign data, customer demographics, or product information could be integrated. This would provide insights into how marketing activities influence sales, how different customer groups contribute to revenue, and which products perform best in different regions, leading to more informed business decisions.

RESULT:

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in both Tableau and Power BI. By creating relationships between the sources, a unified data model was built. The resulting comprehensive dashboard visualized sales performance against set targets, enabling variance analysis and providing deeper, context-rich insights into regional performance that were not possible with a single data source.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		